

GEORGIOS SYROS

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Education

Northeastern University, Boston, MA, United States 2023 - Present
Ph.D. in Computer Science, **GPA: 4.0**
Advisors: Prof. *Alina Oprea*, Prof. *Cristina Nita-Rotaru*

Northeastern University, Boston, MA, United States 2023 - 2025
B.Sc. in Computer Science, **GPA: 4.0**

Athens University of Economics and Business, Athens, Greece 2019 - 2023
B.Sc. in Computer Science, **GPA: 9.35/10**

Research Experience

Microsoft Research & AI Red Team – Microsoft Corporation, Redmond, WA 2026
Research Intern
Research on security for agentic AI systems.

Cybersecurity & Privacy Institute – Northeastern University, Boston, MA 2023 - Present
Graduate Research Assistant
Research on security for systems that employ machine learning and AI.

Maryland Cybersecurity Center – University of Maryland, College Park, MD 2022
Undergraduate Research Assistant
Research on security for machine learning in database management systems.

Mobile Multimedia Lab – Athens University of Economics and Business, Greece 2021 - 2023
Undergraduate Research Assistant
Research on distributed ledger technologies and decentralized storage networks.

Publications

MUZZLE: Adaptive Agentic Red-Teaming of Web Agents Against Indirect Prompt Injection Attacks, USENIX Security, 2026
Georgios Syros, Evan Rose, Brian Grinstead, Christoph Kerschbaumer, William Robertson, Cristina Nita-Rotaru, Alina Oprea,
An automated prompt injection vulnerability discovery framework for AI web agents.

SAGA: A Security Architecture for Governing AI Agentic Systems, NDSS, 2026
Georgios Syros, Anshuman Suri, Jacob Ginesin, Cristina Nita-Rotaru, Alina Oprea,
A secure architectural framework to control communication and capabilities of AI agents.

DROP: Poison Dilution via Knowledge Distillation for Federated Learning, preprint, 2025
Georgios Syros*, Anshuman Suri*, Farinaz Koushanfar, Cristina Nita-Rotaru, Alina Oprea,
A defense that mitigates backdoor attacks in FL by distilling clean knowledge from poisoned models.

Backdoor Attacks in Peer-to-Peer Federated Learning, ACM Transactions on Privacy and Security (TOPS), 2024

Georgios Syros*, Gokberk Yar*, Simona Boboila, Cristina Nita-Rotaru, Alina Oprea,
Analyzes and demonstrates novel backdoor attacks in decentralized federated learning settings.

Decentralized NFT-based Evolvable Games, 4th Conference on Blockchain Research and Applications for Innovative Networks and Services (BRAINS), 2022

Christos Karapapas, **Georgios Syros**, Iakovos Pittaras, George C. Polyzos,
Proposes a blockchain-based framework designed to support artist sustainability in digital games, enabling the use of NFTs as dynamic, evolvable assets that respond to player actions.

Talks

MUZZLE: Adaptive Agentic Red-Teaming of Web Agents Against Indirect Prompt Injection Attacks

@ *Khoury Security and Privacy Day, Northeastern University, Boston,* Apr 2026

Toward Automated Holistic Red-Teaming of Web Agents

@ *New England Security Day, UMass, Amherst,* Mar 2026

SAGA: A Security Architecture for Governing AI Agentic Systems

@ *Network and Distributed System Security (NDSS) Symposium 2026, San Diego, CA,* Feb 2026

From Autonomy to Accountability: Securing Agentic AI Systems with SAGA

@ *Khoury Security Day '25,* May 2025

DROP: Poison Dilution via Knowledge Distillation for Federated Learning

@ *New England Systems Day '25,* Feb 2025

Teaching Experience

Northeastern University

Assisted students with their projects and graded assignments for the following courses:

- CS3650 Computer Systems Fall 2025
- CS6620 Fundamentals of Cloud Computing Summer 2025

Awards and Honors

- Financial Scholarship for Academic Excellence, *Huawei Hellas Enterprise*
- Honor for High Academic Performance, *Athens University of Economics and Business*
- Selected for Greek delegation, *Huawei Seeds for the Future 2021*

Academic Service

Served on the Program Committee:

- ACM Conference on AI and Agentic Systems (CAIS)

Reviewed submissions for:

- ACM Transactions on Privacy and Security (TOPS) Journal
- ACM Conference on Computer and Communications Security (CCS)

* Equal contribution.